

Advanced Fine-tuning and Model Evaluation - Class Notes

Part 1: Advanced Fine-Tuning Techniques

- PEFT and LoRA allow task-specific adaptation with fewer parameters.
- Ideal for resource-constrained environments.

Part 2: Prompt Engineering

- Prompts guide the model's response.
- Changing structure and wording affects output quality.

Part 3: Decoding Strategies

- Beam Search: Deterministic, high coherence.
- Nucleus Sampling: Probabilistic, more diverse results.

Part 4: Conditional Text Generation

- Generates text based on specific input or context.
- Used in translation, summarization, chatbots.

Part 5: Multimodal Integration

- Speech-to-Text with SpeechRecognition
- Text-to-Speech with gTTS

Part 6: Evaluation Metrics

- BLEU: Measures n-gram overlap, good for translation.
- ROUGE: Focuses on recall, good for summarization.
- METEOR: Synonym matching, grammar-aware.
- BERTScore: Embedding-based, measures semantic similarity.

Part 7: Qualitative and Human Evaluation

- Fluency, originality, factual accuracy, bias detection.
- Human judgment is essential for nuanced quality checks.

Homework

1. Fine-tune FLAN-T5 using PEFT/LoRA.
2. Test different prompts with GPT2.
3. Evaluate with BLEU, ROUGE, BERTScore.
4. Write qualitative analysis of model outputs.